W5_Statistical_Analysis

2026-02-06

The objective of today class is to learn how to perform multiple methods of statistical analysis using R. We are going to start with clusters, follow by t-test, ANOVA, and regression analysis. Remember that this is just a subset of the statistical analysis that you can use for your project. Please let me know if there is a topic you are interesting in learning and I can bring an example to class!

We are going to work with a data set on American University Data. The Integrated Postsecondary Education Data System (IPEDS) is the primary source for data on colleges, universities, and technical and vocational postsecondary institutions in the United States (data from https://public.tableau.com/app/learn/sample-data). You can find more information on National Center for Education Statistics in the following link: https://nces.ed.gov/ipeds/.

In this data set, each row represents a university and we have information on multiple variables such as:

- University name, zip code, state, and county
- Geographical region: states grouped by region
- Highest degree offered
- Number of Applicants, Number of Admissions, and Number of Enrolled students
- SAT/ACT score information
- Tuition and fees
- Control of institution (public or private)
- Enrollment by graduate and undergraduate
- Percent of total enrollment by race
- Percent of undergrad enrollment by race
- Financial aid

Each row in this data set represents an academic institution and the unique identifier is the ID of the institution (not the name).

First we are going to activate the library and read the data we are going to be using:

Packages:

```
#install.packages("kableExtra")
library(tidyverse)
library(ggplot2)
library(shiny)
library(plotly)
library(dplyr)
library(readxl)
library(kableExtra)
```

Data:

The file we are going to use is called "IPEDS_Data.xslx". The Zip code format is a 9 digit code for some universities and a 5 digit for other institutions. I am going to clean the data for all universities to have a 5 digit Zip code.

```
ipeds=read_xlsx("IPEDS_Data.xlsx", sheet="Data")

#Clean the zip code
ipeds=ipeds%>%
  separate(ZIP, into="ZIP" , sep = "-", extra="drop")

#extra drops any values after the - because I am only creating one column
```

Note that in the code above, extra='drop' drops any values after the separator '-'. In this case I want to drop the information after the "-" because I am only interested in the first five digits.

Analyzing the data

Let's create a summary of the numerical variables:

```
#for some reason nothing is saving correctly i am so sorry
#Select the numeric variables
num_ipeds = ipeds %>%
  select(where(is.numeric),-ID,-ZIP,-Longitude,-Latitude)

# Find the mean, min, max, and median of each variable
num_summary=num_ipeds%>%
  summarise(across(
    everything(),
    list(
      Mean=~round(mean(.,na.rm=TRUE),2),
      Min = ~ min(., na.rm = TRUE),
      Max = ~ max(., na.rm = TRUE),
      Median=~ median(.,na.rm=TRUE)
    )
  ))

#Formatting the data
num_summary=num_summary%>%
  pivot_longer(
      cols = everything(),
     names_to = c("Variable", ".value"),
     names_pattern = "(.+)_(Mean|Median|Min|Max)$"
    )

# Printing the summary
num_summary
```

```
## # A tibble: 44 x 5
##    Variable     Mean   Min   Max Median
##    <chr>       <dbl> <dbl> <dbl>  <dbl>
##  1 Applicants 7299.    119 72676  3672.
```

```
##  2 Admissions 4027.     62 35815  2242
##  3 Enrolled   1182.     33 10241   590
##  4 SAT          55.6     1    99    66
##  5 ACT          53.9     1    99    49
##  6 SAT_R25     472.    310   720   460
##  7 SAT_R75     581.    360   800   570
##  8 SAT_M25     483.    280   770   470
##  9 SAT_M75     591.    420   800   580
## 10 ACT_25       20.7    13    33    20
## # i 34 more rows
```

Note that the format of the table below is not great for presentation. The kableExtra library, allows you to format the table. The kable_styling allows you to select different formats.

```r
num_summary %>%
  kbl(
    caption = "Summary statistics for numeric variables",
    booktabs = TRUE
  ) %>%
  kable_styling()
```

Now, let's create a summary of categorical variables

```r
# Select all categorical variables
cat_ipeds = ipeds %>%
  select(where(~is.character(.)))

# Select each category (variable), count the number of unique values
# We also provide an example of the categories (at  most 3)
cat_summary=cat_ipeds%>%
  summarise(across(
    everything(),
    list(
      Categories=~n_distinct(.),
      Example=~paste(head(unique(na.omit(.)),3),collapse=", ")
    )
  ))

#Fix the format
cat_summary=cat_summary%>%
  pivot_longer(
      cols = everything(),
     names_to = c("Variable", ".value"),
     names_pattern = "(.+)_(Categories|Example)$"
    )

# Nicer format
cat_summary %>%
  kbl(
    caption = "Summary statistics for categorical variables",
    booktabs = TRUE
  ) %>%
  kable_styling(full_width = FALSE) %>%
  column_spec(3, width = "10cm")
```

Table 1: Summary statistics for numeric variables

Variable	Mean	Min	Max	Median
Applicants	7298.68	119	72676	3672.5
Admissions	4026.58	62	35815	2242.0
Enrolled	1181.77	33	10241	590.0
SAT	55.55	1	99	66.0
ACT	53.90	1	99	49.0
SAT_R25	472.28	310	720	460.0
SAT_R75	580.95	360	800	570.0
SAT_M25	482.65	280	770	470.0
SAT_M75	590.75	420	800	580.0
ACT_25	20.71	13	33	20.0
ACT_75	25.71	14	35	25.0
Num_Bachelors	1274.67	16	12080	585.5
Num_Master	509.83	0	6958	221.0
Num_Doctor	123.54	0	1964	0.0
Tuition_2011	19109.71	3360	43304	20047.0
Tuition_2012	20080.41	3744	45290	21025.0
Tuition_2013	20939.02	3770	47246	21818.0
Tuition_2014	21711.67	3850	49138	22715.0
Un_enrollment	6400.62	176	51333	2914.5
Gr_enrollment	1929.36	0	29874	708.5
American_Indian_or_Alaska_Native	0.50	0	26	0.0
Asian	4.07	0	39	2.0
Black_or_African_American	12.41	0	100	7.0
Hispanic_Latino	8.48	0	93	5.0
Native_Hawaiian_or_other_pacific_islander	0.09	0	14	0.0
White	61.29	0	96	66.0
Two_or_more	2.50	0	28	2.0
Unknown	5.60	0	62	4.0
Nonresident	4.81	0	46	3.0
Asian_Native_Hawaiian_Pacific_Islander	4.27	0	41	2.0
Women	57.19	3	100	57.0
GR_4	39.91	0	93	37.0
GR_5	52.86	2	95	52.0
GR_6	56.20	3	97	55.0
Financial_aid	90.89	41	100	96.0
Federal_state_local_institutional_aid	84.90	25	100	93.0
Federal_grant_aid	38.12	5	96	37.0
Pell_grants	37.44	5	94	36.0
Other_federal_grant_aid	14.29	0	83	11.0
State_Local_aid	36.39	0	96	35.0
Insititutional_grant_aid	73.91	1	100	87.0
Student_loan_aid	64.48	3	100	67.0
Federal_students_loans	63.72	0	100	66.0
Other_loan_aid	7.82	0	83	6.0

Table 2: Summary statistics for categorical variables

Variable	Categories	Example
Name	976	Alabama A & M University, University of Alabama at Birmingham, University of Alabama in Huntsville
ZIP	943	35762, 35294, 35899
Degree	6	Doctor's degree - research/scholarship, Doctor's degree - research/scholarship and professional practice, Bachelor's degree
County	498	Madison County, Jefferson County, Montgomery County
Masters	2	Yes, Implied no
Doctorate	2	Yes, Implied no
State	51	Alabama, Alaska, Arizona
Geo_region	8	Southeast AL AR FL GA KY LA MS NC SC TN VA WV, Far West AK CA HI NV OR WA, Southwest AZ NM OK TX
Control	2	Public, Private not-for-profit
Hist_Black	2	Yes, No

column_spec(n,width=), specified the width in cm of the nth column.

Cluster analysis

We are going to cluster the data by SAT, Tuition 2013 and Percentage of women. We first need to create a new data frame with the variables we are going to use. It is important to create a new data frame because the cluster function requieres a table as input parameter. Remember to pick **numerical** variables.

```
clusterData=ipeds%>%
  select(SAT, Tuition_2013, Women

  )
```

Now, we are going to change the name of the rows to the ID of the institution to keep track of the institution in each row and later join it to the ipeds table.

```
rownames(clusterData)=ipeds$ID
```

```
## Warning: Setting row names on a tibble is deprecated.
```

Now, we can create the clusters using the kmeans function. We are defining an object of type cluster called kclusters where we are going to save the information of the clusters. Don't forget to decide on the number of clusters (k). In this case, we are doing 3 clusters.

```
#I do not know what is going on with my code right now?
kclusters=kmeans(clusterData,centers=2)
```

Note that if you print the object clusters you get a lot of information. If you want to access a particular item, you can use the following available components:

- [1] cluster
- [2] centers

- [5] tot.withinss (for the elbow method)
- [7] sizes of each cluster

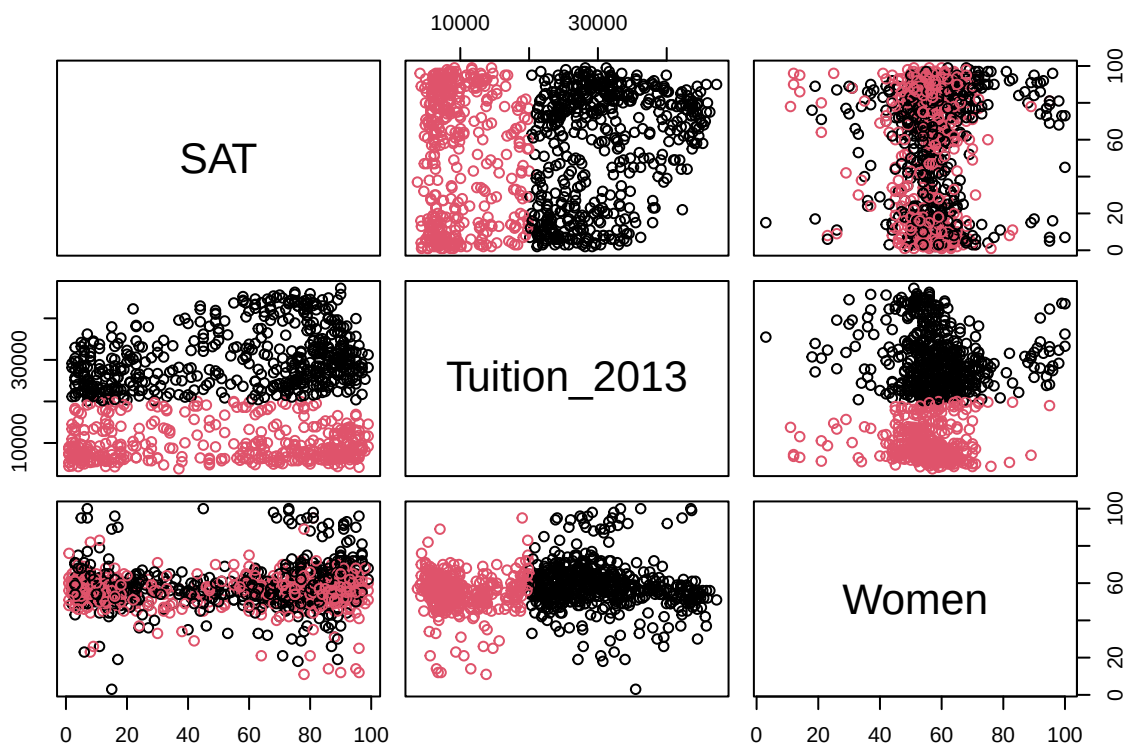
For example, if we want the sizes of each cluster we can do:

```
kclusters[2]
```

```
## $centers
##        SAT Tuition_2013    Women
## 1 57.42322    30401.212 58.66105
## 2 53.31696     9660.422 55.43080
```

Now, let's plot each cluster. Each dot is an institution and each color is a cluster.

```
#YAY i fixed it phew
plot(clusterData,col=kclusters$cluster)
```



Standarizing the data

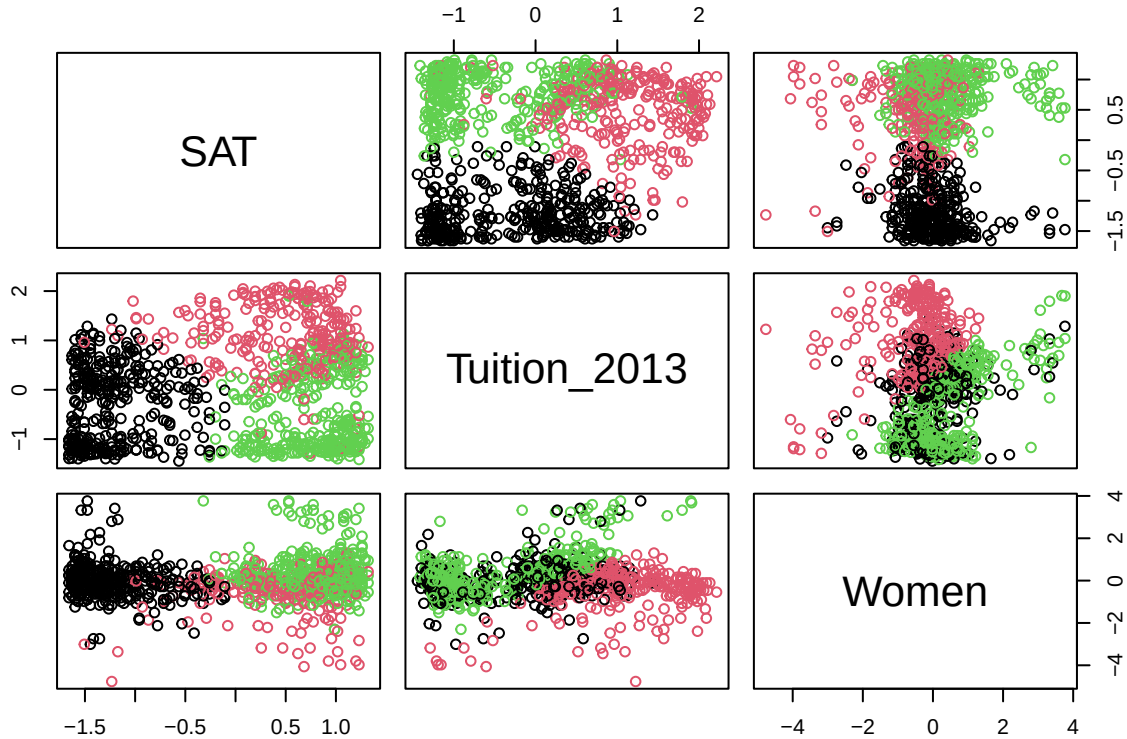
If the variables you use for the clusters have different ranges, it will affect the results. Tuition has a wider range than the percentage of women and SAT percentage. Thus, tuition will have a higher weight on distance than the other two variables. To solve this, we standardize the data.

For that, we use the function scale() as below. We are also going to make sure that the resulting object is a data frame by using the function as.data.frame().

```
scale_data=as.data.frame(scale(clusterData))
```

Now, we can repeat the analysis as we did before but this time with the scaled data. Look at the difference of the resulting clusters.

```
kclusters=kmeans(scale_data,centers=3)
plot(scale_data ,col=kclusters$cluster )
```




```
#nice colors good good easier to read and visualize clusters
```

Now, we want to save the information of clusters as a new column in the data frame with all the information of the institutions. We are going to use the names of the rows to create a new data frame with the ID of the institution and the cluster. We can then join it with the ipeds data. I am going to create a new data frame called ipeds_cluster with all the information of the institutions because I don't want to overwrite the ipeds data.

```
clus_final=as.data.frame(kclusters$cluster)
clus_final=tibble(ID=row.names(clus_final) ,Cluster= as.character(clus_final))

ipeds_cluster=merge(ipeds,clus_final,by="ID")
```

Now you can plot or summarize any of the variables and color the dots depending on the cluster. For example, if we want to find the average SAT, percentage of women, and tuition value for each of the clusters, we can do the following:

```r
Cluster_centers=ipeds_cluster%>%
  group_by(Cluster)%>%
  summarise(SAT= round(mean(SAT),2),
            Women= round(mean(Women),2) ,
            Tuition=round(mean(Tuition_2013),2)
              )

Cluster_centers
```

```
## # A tibble: 1 x 4
##   Cluster                                                    SAT Women Tuition
##   <chr>                                                    <dbl> <dbl>   <dbl>
## 1 "c(1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, ~  55.6  57.2  20939.
```

Note that the previous information are the cluster centers.

You can also plot these variables with every color representing a cluster. We are going to do a 3D plot to see all the variables together.

Note that 3D plots **can not** be knitted to a PDF document. However you can use it in your Shiny App.

```r
#woah this is super cool i want to do this for a separate project I have... saving this for later!!
plot_ly(x=ipeds_cluster$SAT,y=ipeds_cluster$Women,
        z=ipeds_cluster$Tuition_2013,color=ipeds_cluster$Cluster) %>%
          add_markers() %>%
           layout(scene = list(xaxis = list(title = 'SAT Percentage'),
                     yaxis = list(title = 'Percentage of Women'),
                     zaxis = list(title = 'Tuition and fees 2013')))
```

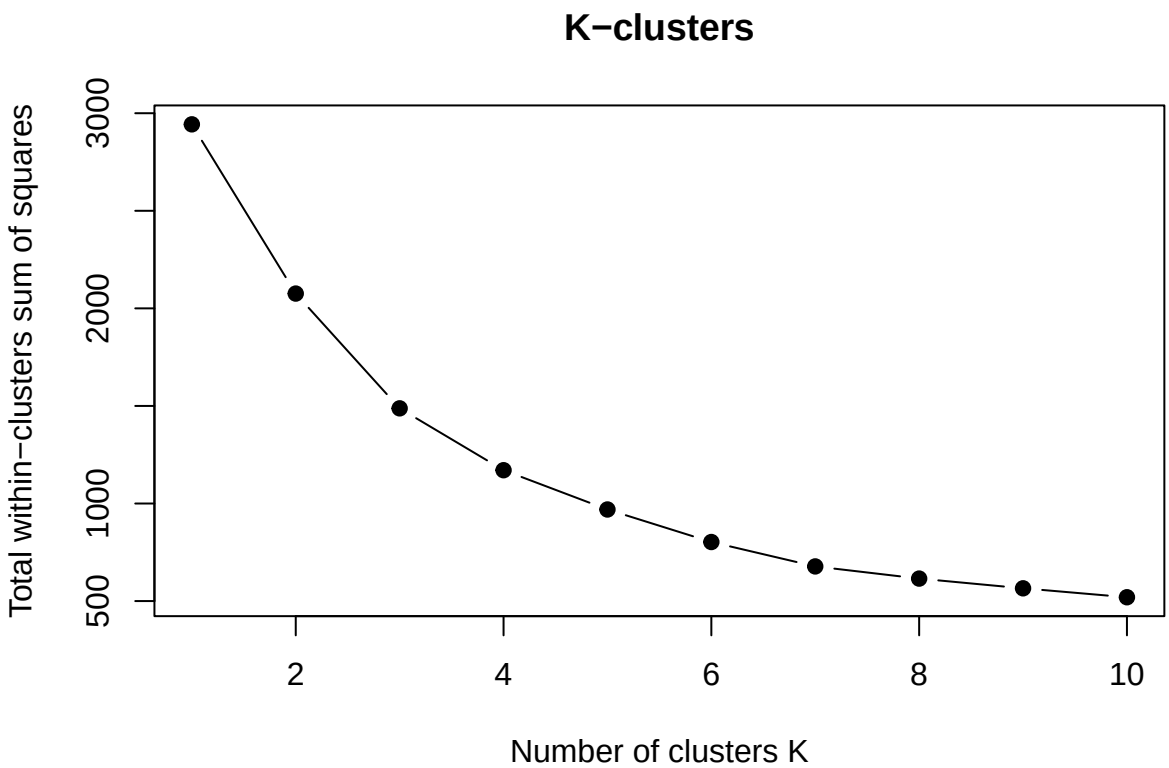
Elbow method

The following code creates the plot for the elbow method. We are going to change the value of k from 1 to 10, but you can adapt the code to any other range.

```r
#Using the scaled data
data=scale_data
k.max = 10
#Create a vector with the within sum of squarer (wss)
wss = sapply(1:k.max,
              function(k){kmeans(data, k, nstart=50,iter.max = 15 )$tot.withinss})
wss
```

```
##  [1] 2943.0000 2075.8265 1487.6094 1170.6906  969.3392  802.6624  677.6328
##  [8]  615.2572  565.4045  519.6451
```

```r
#Plot the wss vs number of iterations
plot(1:k.max, wss,
     type="b", pch=19,
     xlab="Number of clusters K",
     ylab="Total within-clusters sum of squares", main="K-clusters")
```



t-test

t-tests are used to assess the difference between the difference of means (either one sample or two sample). Remember that you reject the null hypothesis in favor of the alternative hypothesis if the p-value is less than alpha.

There are 3 types of t-test:

- One-sample t-test
- Independent sample t-test (pooled and non-pooled)
- Paired t-test

One sample t-test

In the following t-test we are going to test if the percentage of women in the institutions is greater than 50%.

Null: $H_0 : \mu_W = 50\%$ Alternative: $H_0 : \mu_W > 50\%$

You can define the alternative in the function to be "greater", "less", or "two.sided". The default value is two sided (non equal)

```
t.test(ipeds$Women , mu =50 , conf.level = 0.98 ,alternative ="greater" )
```

```
##
##  One Sample t-test
```

```
##
## data:  ipeds$Women
## t = 19.811, df = 981, p-value < 2.2e-16
## alternative hypothesis: true mean is greater than 50
## 98 percent confidence interval:
##  56.4413     Inf
## sample estimates:
## mean of x
##  57.18737
```

Two-sample t-test

Let's now compare the admissions at public and private universities. Our null hypothesis is going to be that the mean is the same, and our alternative that it is different.

Null: $H_0 : \mu_{Pu} = \mu_{Pr}$ Alternative: $\mu_{Pu} \neq \mu_{Pr}$

Let's create a subset with the information of the public institutions and one subset with the data of the private institutions.

```r
public=ipeds[ipeds$Control=="Public",]
private=ipeds[ipeds$Control=="Private not-for-profit",]
```

Non- pooled test We are going to build a non-pooled test (the variance of the groups is not equal) first

```r
t.test(public$Admissions ,private$Admissions ,paired =FALSE  ,var.equal = FALSE,conf.level = 0.95)
```

```
##
##  Welch Two Sample t-test
##
## data:  public$Admissions and private$Admissions
## t = 12.827, df = 479.79, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  3578.551 4873.304
## sample estimates:
## mean of x mean of y
##  6694.682  2468.755
```

Pooled test We can also build a pooled test (the variance of the groups is equal). We are going to save the information of the test in an object call "test:

```r
#why is it not printing
test=t.test(public$Admissions ,private$Admissions ,paired =FALSE ,var.equal = TRUE,conf.level = 0.95, al
```

By saving the information of the test in an object (variable), we can use the object to access different results of the test and incorporate it in a report as text. The test has functions as p-value, statistic (t-statistic value), confidence interval, among others. The following code summarizes the results:

```r
cat("For the test the p-value is ", as.numeric(test$p.value), ", the statistic t value is", round(test$s
```

```
## For the test the p-value is  9.869543e-46 , the statistic t value is 14.9 and the confidence interval
```

```r
cat("The mean of the private universities is ", round(mean(private$Admissions),2), "and the mean of the
```

```
## The mean of the private universities is  2468.75 and the mean of the public universities is  6694.68
```

Paired test Let's now compare if the difference of the tuition from 2012 to 2013 is greater than 1000. This is a paired test because we have one data point for each year for each institution and we want to test on the difference. Our null hypothesis is going to be that the difference is 1000, and our alternative that it is greater.

Null: $H_0 : \doteq 1000$ Alternative: $d > 1000$

```r
t.test(ipeds$Tuition_2013 ,ipeds$Tuition_2012 ,paired = TRUE ,conf.level = 0.95, alternative = "greater
```

```
##
##  Paired t-test
##
## data:  ipeds$Tuition_2013 and ipeds$Tuition_2012
## t = 41.774, df = 981, p-value < 2.2e-16
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
##  824.7666      Inf
## sample estimates:
## mean difference
##        858.6059
```

One-way ANOVA

The one-way ANOVA is a test of equality of means comparing variances. If the variance within groups is statistically smaller than the variance between groups, then, we can reject the null hypothesis and conclude that the mean of at least one group is different to the other groups.

Lets test if the average enrollment by region is different.

```r
anova = aov(Enrolled ~ Geo_region, data = ipeds)

summary(anova)
```

```
##              Df    Sum Sq Mean Sq F value   Pr(>F)
## Geo_region    7 6.189e+07 8841978   4.542 5.33e-05 ***
## Residuals   974 1.896e+09 1946831
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Geo region represent between groups and residuals is within groups.

Regression analysis

In regression analysis we want to study the relationship between two (or more) variables; a dependent variable (VOI - y) and one or more independent variables (predictor(s) -x_i)

Don't forget to look at the following and conclude about your results:

- Coefficient of each independent variable and it's p-value
- R-square of the model

Simple linear regression (one independent variable and one dependent) The following simple linear regressions looks at the 4 year graduation rate (GR_4) as a function of the SAT reading 75th percentile.

```
#ah crap i missed this
#Simple linear regression
model_1 = lm(formula = GR_4 ~SAT_R75, data = ipeds)
summary(model_1)
```

```
##
## Call:
## lm(formula = GR_4 ~ SAT_R75, data = ipeds)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -51.548  -9.730   1.061  10.075  37.492
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -87.569276   3.886300  -22.53   <2e-16 ***
## SAT_R75       0.219433   0.006643   33.03   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.27 on 980 degrees of freedom
## Multiple R-squared:  0.5268, Adjusted R-squared:  0.5263
## F-statistic:  1091 on 1 and 980 DF,  p-value: < 2.2e-16
```

You can access specific information of the results as follows: - $coefficients[1]: intercept - $coefficients[2]: slope

```
cat("The model results show that the slope is ", round(model_1$coefficients[2],2), "and the intercept is
```

```
## The model results show that the slope is  0.22 and the intercept is  -87.57
```

Confidence interval You can find the confidence interval of the estimates of the regression by using the following code:

```
confint(model_1, level=0.95)
```

```
##                  2.5 %      97.5 %
## (Intercept) -95.195704 -79.9428485
## SAT_R75       0.206396   0.2324701
```

Multiple linear regression (two or more independent variables and one dependent) The following linear regressions looks at the 4 year graduation rate (GR_4) as a function of the SAT reading 75th percentile and undergraduate enrollment.

```
model_2 = lm(formula=GR_4 ~ SAT_R75 + Un_enrollment , data = ipeds)
summary(model_2)
```

```
##
## Call:
## lm(formula = GR_4 ~ SAT_R75 + Un_enrollment, data = ipeds)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -54.334  -8.916   0.654   9.228  35.263
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -9.073e+01  3.728e+00  -24.34   <2e-16 ***
## SAT_R75        2.308e-01  6.457e-03   35.75   <2e-16 ***
## Un_enrollment -5.412e-04  5.580e-05   -9.70   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.64 on 979 degrees of freedom
## Multiple R-squared:  0.5683, Adjusted R-squared:  0.5674
## F-statistic: 644.3 on 2 and 979 DF,  p-value: < 2.2e-16
```

You can also include categorical variables. The following linear regressions looks at the 4 year graduation rate (GR_4) as a function of the SAT reading 75th percentile and control (if the institution is public or private).

```r
#ah crap i missed this... noooooo i think my headache is getting worse regardless of how much water i d
model_3 = lm(GR_4~ SAT_R75 +Control,  data = ipeds)
summary(model_3)
```

```
##
## Call:
## lm(formula = GR_4 ~ SAT_R75 + Control, data = ipeds)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -45.816  -8.576   0.837   8.547  32.619
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -75.69944    3.55081  -21.32   <2e-16 ***
## SAT_R75         0.20753    0.00598   34.70   <2e-16 ***
## ControlPublic -13.43483    0.84988  -15.81   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.75 on 979 degrees of freedom
## Multiple R-squared:  0.623,  Adjusted R-squared:  0.6222
## F-statistic:   809 on 2 and 979 DF,  p-value: < 2.2e-16
```

Usually R takes the first category (alphabetically) as the reference category. If you want to change the reference category you can use ref=. First we have to convert the variable to a factor. The following linear regressions looks at the 4 year graduation rate (GR_4) as a function of the SAT reading 75th percentile and the highest degree offered.

Create the model:

```r
model_4 = lm(GR_4 ~ SAT_R75 + Degree ,  data = ipeds)
summary(model_4)
```

```
##
## Call:
## lm(formula = GR_4 ~ SAT_R75 + Degree, data = ipeds)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -54.721  -9.326   1.042   9.871  33.946
##
## Coefficients:
##                                                                  Estimate
## (Intercept)                                                    -84.403730
## SAT_R75                                                          0.225246
## DegreeDoctor's degree -  professional practice                  -3.916585
## DegreeDoctor's degree - other                                   -9.041028
## DegreeDoctor's degree - research/scholarship                    -9.799633
## DegreeDoctor's degree - research/scholarship and professional practice  -9.538006
## DegreeMaster's degree                                           -5.397822
##                                                                Std. Error
## (Intercept)                                                      4.215502
## SAT_R75                                                          0.006815
## DegreeDoctor's degree -  professional practice                   2.050940
## DegreeDoctor's degree - other                                    3.426520
## DegreeDoctor's degree - research/scholarship                     1.804755
## DegreeDoctor's degree - research/scholarship and professional practice   1.651181
## DegreeMaster's degree                                            1.576343
##                                                                t value
## (Intercept)                                                    -20.022
## SAT_R75                                                         33.052
## DegreeDoctor's degree -  professional practice                  -1.910
## DegreeDoctor's degree - other                                   -2.639
## DegreeDoctor's degree - research/scholarship                    -5.430
## DegreeDoctor's degree - research/scholarship and professional practice  -5.776
## DegreeMaster's degree                                           -3.424
##                                                                Pr(>|t|)
## (Intercept)                                                     < 2e-16
## SAT_R75                                                         < 2e-16
## DegreeDoctor's degree -  professional practice                 0.056471
## DegreeDoctor's degree - other                                  0.008459
## DegreeDoctor's degree - research/scholarship                   7.12e-08
## DegreeDoctor's degree - research/scholarship and professional practice 1.03e-08
## DegreeMaster's degree                                          0.000642
##
## (Intercept)                                                    ***
## SAT_R75                                                        ***
## DegreeDoctor's degree -  professional practice                 .
## DegreeDoctor's degree - other                                  **
## DegreeDoctor's degree - research/scholarship                   ***
## DegreeDoctor's degree - research/scholarship and professional practice ***
## DegreeMaster's degree                                          ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.97 on 975 degrees of freedom
## Multiple R-squared:  0.5487, Adjusted R-squared:  0.546
## F-statistic: 197.6 on 6 and 975 DF,  p-value: < 2.2e-16
```

Change the reference category to "Master's degree" and change the model:

```r
ipeds$Degree=as.factor(ipeds$Degree )
ipeds$Degree = relevel(ipeds$Degree , ref = "Master's degree" )
model_5 = lm(GR_4 ~ SAT_R75+Degree,  data = ipeds)
summary(model_5)
```

```
##
## Call:
## lm(formula = GR_4 ~ SAT_R75 + Degree, data = ipeds)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -54.721  -9.326   1.042   9.871  33.946
##
## Coefficients:
##                                                                  Estimate
## (Intercept)                                                    -89.801552
## SAT_R75                                                          0.225246
## DegreeBachelor's degree                                          5.397822
## DegreeDoctor's degree -  professional practice                   1.481237
## DegreeDoctor's degree - other                                   -3.643206
## DegreeDoctor's degree - research/scholarship                    -4.401811
## DegreeDoctor's degree - research/scholarship and professional practice  -4.140184
##                                                                Std. Error
## (Intercept)                                                      3.903778
## SAT_R75                                                          0.006815
## DegreeBachelor's degree                                          1.576343
## DegreeDoctor's degree -  professional practice                   1.676424
## DegreeDoctor's degree - other                                    3.209130
## DegreeDoctor's degree - research/scholarship                     1.362417
## DegreeDoctor's degree - research/scholarship and professional practice   1.183873
##                                                                t value
## (Intercept)                                                    -23.004
## SAT_R75                                                         33.052
## DegreeBachelor's degree                                          3.424
## DegreeDoctor's degree -  professional practice                   0.884
## DegreeDoctor's degree - other                                   -1.135
## DegreeDoctor's degree - research/scholarship                    -3.231
## DegreeDoctor's degree - research/scholarship and professional practice  -3.497
##                                                                Pr(>|t|)
## (Intercept)                                                     < 2e-16
## SAT_R75                                                         < 2e-16
## DegreeBachelor's degree                                        0.000642
## DegreeDoctor's degree -  professional practice                 0.377147
## DegreeDoctor's degree - other                                  0.256544
## DegreeDoctor's degree - research/scholarship                   0.001275
```

```
## DegreeDoctor's degree - research/scholarship and professional practice 0.000491
##
## (Intercept)                                                          ***
## SAT_R75                                                              ***
## DegreeBachelor's degree                                              ***
## DegreeDoctor's degree -  professional practice
## DegreeDoctor's degree - other
## DegreeDoctor's degree - research/scholarship                         **
## DegreeDoctor's degree - research/scholarship and professional practice ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.97 on 975 degrees of freedom
## Multiple R-squared:  0.5487, Adjusted R-squared:  0.546
## F-statistic: 197.6 on 6 and 975 DF,  p-value: < 2.2e-16
```

Prediction

You can use your models for prediction. We will use model 1 and predict the graduation rate for a school with SAT of 500.

```r
predict(model_1 ,data.frame(SAT_R75=500) )
```

```
##        1
## 22.14726
```

If you want to predict multiple values, you can create a vector with the values you want to predict.

```r
#okay i guess we did not have enough time for this yaayyyy
predict(model_1,newdata= data.frame(SAT_R75 = 500))
```

```
##        1
## 22.14726
```

Lastly, you can decide if you want the prediction or the confidence interval for your values.

```r
#Prediction interval
predict(model_1,data.frame(SAT_R75=c(400,550,800)),interval= "confidence", level = 0.95 )
```

```
##         fit       lwr       upr
## 1  0.203951 -2.318757  2.726659
## 2 33.118911 32.138231 34.099591
## 3 87.977178 84.984805 90.969551
```

```r
#Confidence interval
predict(model_1,data.frame(SAT_R75=c(400,550,800)),interval="prediction", level = 0.95)
```

```
##         fit        lwr       upr
## 1  0.203951 -27.919099  28.32700
## 2 33.118911   5.092074  61.14575
## 3 87.977178  59.808114 116.14624
```

Additional information: Step function

The step function calculates the AIC for each variables and based on that criteria, decides which variables are the best one to keep in the model.

The AIC penalizes for adding more variables to the model. The model with lower AIC is preferable.

```
#Create subset of the data with some variables
ipeds_small=ipeds%>%
  select(
    Applicants,
    Admissions,
    Enrolled,
    GR_4,
    SAT_R75,
    SAT_M75,
    Tuition_2014,
    Control,
    Doctorate,
    Num_Bachelors,
    Num_Doctor,
    Num_Master
  )

#Using ., you are telling R to include all the variables in the data
model_2=lm(GR_4~.,data=ipeds_small)

#The step function finds the "best" model using the AIC criteria
model_3=step(model_2)
```

```
## Start:  AIC=4707.84
## GR_4 ~ Applicants + Admissions + Enrolled + SAT_R75 + SAT_M75 +
##     Tuition_2014 + Control + Doctorate + Num_Bachelors + Num_Doctor +
##     Num_Master
##
##                 Df Sum of Sq    RSS    AIC
## - Num_Bachelors  1      19.6 115780 4706.0
## - Num_Doctor     1     217.9 115979 4707.7
## <none>                       115761 4707.8
## - Admissions     1     351.3 116112 4708.8
## - Enrolled       1     715.8 116477 4711.9
## - SAT_M75        1     815.7 116576 4712.7
## - Applicants     1    1345.1 117106 4717.2
## - Control        1    1606.1 117367 4719.4
## - Doctorate      1    2274.7 118035 4724.9
## - Num_Master     1    2456.5 118217 4726.5
## - SAT_R75        1    5199.8 120961 4749.0
## - Tuition_2014   1   26898.5 142659 4911.0
##
## Step:  AIC=4706
## GR_4 ~ Applicants + Admissions + Enrolled + SAT_R75 + SAT_M75 +
##     Tuition_2014 + Control + Doctorate + Num_Doctor + Num_Master
##
##                Df Sum of Sq    RSS    AIC
```

```
## - Num_Doctor    1     233.5 116014 4706.0
## <none>                      115780 4706.0
## - Admissions    1     346.0 116126 4706.9
## - SAT_M75       1     815.4 116596 4710.9
## - Enrolled      1    1109.7 116890 4713.4
## - Applicants    1    1332.0 117112 4715.2
## - Control       1    1633.1 117413 4717.8
## - Doctorate     1    2261.2 118042 4723.0
## - Num_Master    1    2787.7 118568 4727.4
## - SAT_R75       1    5185.5 120966 4747.0
## - Tuition_2014  1   27839.2 143620 4915.6
##
## Step:  AIC=4705.98
## GR_4 ~ Applicants + Admissions + Enrolled + SAT_R75 + SAT_M75 +
##     Tuition_2014 + Control + Doctorate + Num_Master
##
##                Df Sum of Sq    RSS    AIC
## <none>                      116014 4706.0
## - Admissions    1     454.9 116469 4707.8
## - SAT_M75       1     916.2 116930 4711.7
## - Enrolled      1    1474.3 117488 4716.4
## - Control       1    1539.7 117554 4716.9
## - Applicants    1    1620.9 117635 4717.6
## - Doctorate     1    2132.8 118147 4721.9
## - Num_Master    1    2617.0 118631 4725.9
## - SAT_R75       1    5116.7 121131 4746.4
## - Tuition_2014  1   27876.5 143890 4915.4
```